



St Catherine's School

Year: 12
 Subject: 3 Unit Mathematics
 Time Allowed: 2 hours (plus 5 mins reading time)
 Date: August 2000

Exam number: _____

Directions to candidates:

- All questions are to be attempted.
- All questions are of equal value.
- All necessary **working** must be shown in every question.
- Full marks may not be awarded for careless or badly arranged work.
- Each question attempted should be started on a new page.
- Approved calculators and geometrical instruments are required.
- Attach the question paper to the front of **Section A**.
- Write a cover page for **Section B and C** and include your number.
- Hand in your work in 3 bundles:
 - Section A Questions 1, 2 and 3.
 - Section B Questions 4 and 5
 - Section C Questions 6 and 7.

TEACHER'S USE ONLY	
Total Marks	
A	
B	
C	
TOTAL	

Section A

Question 1

- Differentiate $e^{2x} \sin x$
- Find the acute angle between the lines $2x + y = 4$ and $x - y = 2$
- A committee of 3 men and 4 women is to be formed from a group of 8 men and 6 women. Write an expression for the number of ways this can be done.
- Evaluate $\int_0^2 \frac{dx}{4 + x^2}$
- Using the substitution $u = 2x + 1$ or otherwise, find $\int_0^1 \frac{4x}{2x+1} dx$

Question 2

- A particle is moving in simple harmonic motion. It's displacement, x , at time, t , is given by $x = 3 \sin(4t + \frac{\pi}{4})$.
 - find the period and amplitude of the motion
 - find the velocity of the particle when $t = 0$.
 - find the maximum acceleration of the particle.
 - find the speed of the particle when $x = 2$
- The polynomial $P(x) = x^3 + bx^2 + cx + d$ has roots at 0, 3 and -3.
 - find b , c and d
 - without using calculus, sketch the graph of $y = P(x)$
 - Hence or otherwise solve the inequality $\frac{x^2 - 9}{x} \geq 0$.

Question 3

- a) i) Find $\frac{dy}{dx}$ if $y = \tan^{-1}(\sin x)$ 2
- ii) Evaluate $\int_0^1 \frac{dx}{\sqrt{2-x^2}}$ 2
- b) A cup of hot coffee at temperature T degrees Celsius loses heat when placed in a cooler environment. It cools according to the law $\frac{dT}{dt} = k(T - T_0)$ where time, t is the time elapsed in minutes and T_0 is the temperature of the environment in degrees Celsius.
- i) A cup of coffee at 100°C is placed in an environment at -20°C for 3 minutes and then cools to 70°C . Find k . 2
- ii) The same cup of coffee at 70°C is then placed in an environment at 20°C assuming k stays the same, find the temperature of the coffee after a further 15 minutes. 3
- c) The function $h(x)$ is given by $h(x) = \sin^{-1}x + \cos^{-1}x$ for $-1 \leq x \leq 1$.
- i) show that $h'(x) = 0$ 1
- ii) sketch the graph of $y = h(x)$ 2

SECTION B (Start a new page)

Question 4

- a) A spherical balloon is expanding so that its volume is increasing at the constant rate of 10mm^3 per second. What is the rate of increase of the radius when the surface area is 500mm^2 .
($V = \frac{4}{3}\pi r^3$ $SA = 4\pi r^2$) 4
- b) Find the constant term in the expansion of $(3x^2 - \frac{1}{2x})^9$. 4
- c) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. The equation of chord PQ is given by $y - ap^2 = \frac{p+q}{2}(x - 2ap)$.
- i) If PQ is a focal chord show that $pq = -1$ 1
- ii) Find M, the midpoint of PQ. 1
- iii) Find the equation of the locus of M 2

Question 5

A dangerous fire is burning in a low open tank on horizontal ground. Fire fighters are forced to stay 60m away from the fire. They are using a pump which is on the ground and can eject water at 30m/s at any angle to the horizontal, α .

(Assume that $g = 10\text{m/s}^2$ and that all frictional forces, including air resistance, can be neglected.)

- a) Show that the expression for the vertical motion is $y = -5t^2 + 30t \sin \alpha$ $1\frac{1}{2}$
- b) Show that the expression for horizontal motion is $x = 30t \cos \alpha$ $1\frac{1}{2}$
- c) Show that the range of the projectile is given by $x = 90 \sin 2\alpha$ 2
- d) Find the maximum horizontal distance the pump can reach. 1
- e) Find the angle of projection needed for the pumped water to reach the fire. 3
- f) Another other pump is on a vertical stand 5m high and can eject water at 40m/s but only horizontally. Can this pump reach the fire? Justify your answer. 3
(You may use the formulas for the horizontal distance; $x = Vt \cos \alpha$ and vertical distance $y = -\frac{1}{2}gt^2 + Vt \sin \alpha$, where V is the initial velocity and α is the angle of projection and $g=10\text{m/s}^2$).

SECTION C (Start a new page)

Question 6

- a) i) Show that $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^2 x \, dx = \frac{\pi}{8} - \frac{1}{4}$ 2
- ii) The function $g(x)$ is given by $g(x) = 2 + \cos x$. The graph $y = g(x)$ for $\frac{\pi}{4} \leq x \leq \frac{\pi}{2}$ is rotated about the x axis. Find the volume of the solid generated. (You may use the result of a(i)). Give your answer in exact form. 3
- b) The velocity of a point moving along the x axis is given by $v^2 = 16x - 4x^2 + 20$.
- i) Show that $\ddot{x} = -4(x - 2)$ 2
- ii) State the centre of motion 1
- iii) What is the amplitude of the motion 2
- iv) What is the period of the motion 1
- v) Find the maximum speed of the particle 1

Question 7

- a) If $(1+x)^n = \sum_{r=0}^n C_r x^r$ show that $\sum_{r=1}^n r C_r = n 2^{n-1}$ 3
- b) Consider the function $f(x) = (x-2)^2 + 1$
- i) Sketch the parabola $y = f(x)$, showing clearly any intercepts with the axes, and the coordinates of its vertex. Use the same scale on both axes. 1.5
- ii) What is the largest domain containing the value $x = 3$, for which the function has an inverse function $f^{-1}(x)$? 1
- iii) Sketch the function $y = f^{-1}(x)$ on the same set of axes as your graph in part(i). Label the two graphs clearly. 1.5
- (iv) What is the domain of the inverse function? 1
- (v) Let a be a real number not in the domain found in part (ii). Find $f^{-1}[f(a)]$. 2
- (vi) Find the x coordinate of any points of intersection of the two curves $y = f(x)$ and $y = f^{-1}(x)$. 2

END OF EXAMINATION

Table of Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Year in ...

from A

in 1

$$e^{2x} \cdot \sin x =$$

$$e^{2x} \cdot \cos x + \sin x \cdot 2e^{2x}$$

$$e^{2x} (\cos x + 2 \sin x) \quad (2)$$

$$2x + y = 4 \rightarrow y = 4 - 2x$$

$$m_1 = -2$$

$$x - y = 2 \rightarrow y = x - 2$$

$$m_2 = 1$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{-2 - 1}{1 + (-2)(1)} \right|$$

$$= 3$$

$$\theta = 72^\circ \text{ (nearest degree)} \quad (2)$$

Choose 3 from 8 and 4 from 6

$$\therefore \text{No of ways} = {}^8C_3 \times {}^6C_4 \quad (1)$$

$$= 56 \times 15$$

$$= 840$$

$$\int_0^2 \frac{dx}{4+x^2} = \left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^2$$

$$= \frac{1}{2} \tan^{-1} \left(\frac{2}{2} \right) - \frac{1}{2} \tan^{-1} \left(\frac{0}{2} \right)$$

$$= \frac{1}{2} \times \frac{\pi}{4} - \frac{1}{2} \times 0$$

$$= \frac{\pi}{8}$$

(3)

$$e) \int_0^1 \frac{4x}{2x+1} dx$$

$$u = 2x+1 \text{ also } 2x = u-1$$

$$\frac{du}{dx} = 2$$

$$du = 2 \cdot dx$$

$$x=1 \quad u=3$$

$$x=0 \quad u=1$$

$$= \int_1^3 \frac{u-1}{u} du$$

$$= \int_1^3 \left(1 - \frac{1}{u} \right) du$$

$$= \left[u - \log_e u \right]_1^3$$

$$= (3 - \log_e 3) (1 - \log_e 1)$$

$$= (3 - \log_e 3) (1 - 0)$$

$$= 3 - \log_e 3$$

(4)

Question 2

$$a) x = 3 \sin \left(4t + \frac{\pi}{4} \right)$$

$$i) \text{ amplitude} = 3 \quad (1/2)$$

$$\text{period} = \frac{2\pi}{n}$$

$$= \frac{2\pi}{4}$$

$$= \frac{\pi}{2} \quad (1/2)$$

$$ii) v = \frac{dx}{dt}$$

$$= 3 \cos \left(4t + \frac{\pi}{4} \right) \cdot 4$$

$$= 12 \cos \left(4t + \frac{\pi}{4} \right) \quad (1)$$

when $t=0$

$$v = 12 \cos \left(\frac{\pi}{4} \right)$$

$$= 12 \times \frac{1}{\sqrt{2}}$$

$$= 12 \times \frac{\sqrt{2}}{2}$$

$$= 6\sqrt{2} \quad (1)$$

(iii) max acc occurs when $v=0$

$$v = 12 \cos \left(4t + \frac{\pi}{4} \right)$$

$$0 = 12 \cos \left(4t + \frac{\pi}{4} \right)$$

$$4t + \frac{\pi}{4} = \frac{\pi}{2}$$

$$4t = \frac{\pi}{4}$$

$$t = \frac{\pi}{16} \quad (1)$$

$$a = \frac{dv}{dt}$$

$$= -48 \sin \left(4t + \frac{\pi}{4} \right)$$

$$\text{at } t = \frac{\pi}{16}$$

$$a = -48 \sin \left(4 \left(\frac{\pi}{16} \right) + \frac{\pi}{4} \right)$$

$$= -48 \sin \left(\frac{\pi}{2} \right)$$

$$= -48$$

(1)

$\therefore$ max acceleration is -48 m/s^2

$$iv) v^2 = n^2 (a^2 - x^2) \quad n=4 \checkmark$$

$$\text{when } x=2 \quad a=3 \checkmark$$

$$v = \sqrt{40} \checkmark \quad x=2 \checkmark$$

$\therefore$ speed is 180 m/s or 41

$$x^3 + bx^2 + cx + d = x(x+3)(x-3)$$

$$P(0) = 0 \quad 0 = d$$

$$P(3) = 0 \quad 0 = 27 + 9b + 3c$$

$$P(-3) = 0 \quad 0 = -27 + 9b - 3c$$

$$\textcircled{1} + \textcircled{2}$$

$$0 = 18b$$

$$b = 0 \quad \checkmark$$

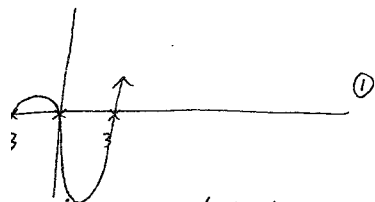
$$\therefore 0 = 27 + 3c$$

$$27 = -3c$$

$$c = -9 \quad \checkmark$$

$$b = 0, c = -9, d = 0 \quad \textcircled{2}$$

$$\text{ii)} \quad y = P(x) = x^3 - 9x$$



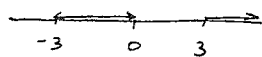
$$\text{at } x = 1 \quad y = -8$$

$$\frac{x^2 - 9}{x} = 0$$

$$x^2 - 9 = 0$$

$$(x-3)(x+3) = 0$$

$x = 3, -3$ are critical points



$$\text{at } x = 1 \quad \frac{x^2 - 9}{x} = -8$$

etc

$$-3 \leq x < 0 \quad \text{or} \quad x \geq 3$$

or

$$x(x^2 - 9) \geq 0$$

$$\frac{9}{5}$$

$$\text{let } u = \tan x$$

$$y = \tan^{-1} u$$

$$\frac{du}{dx} = \cos x$$

$$\frac{dy}{du} = \frac{1}{1+u^2}$$

$$\frac{dy}{dx} = \cos x \times \frac{1}{1+\sin^2 x}$$

$$= \frac{\cos x}{1+\sin^2 x}$$

$\textcircled{2}$

$$\text{ii)} \quad \int_0^1 \frac{du}{\sqrt{2-u^2}} = \left[\sin^{-1} \frac{u}{\sqrt{2}} \right]_0^1$$

$$= \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) - \sin^{-1} \frac{0}{\sqrt{2}}$$

$$= \frac{\pi}{4}$$

$\textcircled{2}$

3b)

$$\frac{dT}{dt} = k(T - T_0) \quad \therefore T = T_0 + Ae^{kt}$$

$$\text{i)} \quad T_0 = -20$$

$$\text{At } t = 0, \quad T = 100$$

$$100 = -20 + A$$

$$A = 120$$

$$\text{at } t = 3 \quad T = 70$$

$$70 = -20 + 120e^{3k}$$

$$e^{3k} = \frac{9}{12}$$

$$k = \frac{1}{3} \ln \frac{3}{4}$$

$$= -0.095894 \dots$$

$\textcircled{2}$

ii) let $t = 0$ when app placed in environment 20°C

$$T = 20 + Be^{kt}$$

$$\text{at } t = 0 \quad T = 70^\circ$$

$$70 = 20 + Be^0$$

$$B = 50$$

$$T = 20 + 50e^{kt}$$

$$\text{where } k \approx 0.095894 \dots$$

$$\text{at } t = 15 \quad T = 20 + 50e^{15k}$$

$$= 31.86^\circ \quad \textcircled{3}$$

3c/

$$h(x) = \sin^{-1} x + \cos^{-1} x \quad 0 \leq x \leq 1$$

$$h'(x) = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1+x^2}}$$

$$= 0$$

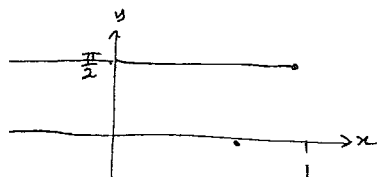
①

$\therefore$ The function $h(x)$ has a gradient of zero

$\therefore$ The fn is a straight line of the form $y = a$ where a is a constant.

$$\text{when } x=0 \quad h(x) = \frac{\pi}{2}$$

$$\therefore h(x) = \frac{\pi}{2} \quad \text{for } 0 \leq x \leq 1$$



Question 4

$$\frac{dV}{dt} = \frac{d}{dt} \left(\frac{4}{3} \pi R^3 \right) \checkmark$$

$$= \frac{d}{dr} \left(\frac{4}{3} \pi r^3 \right) \times \frac{dr}{dt} \checkmark$$

$$= 4\pi R^2 \frac{dr}{dt} \checkmark$$

$$\text{hence } \frac{dV}{dt} = 10 \text{ and } 4\pi R^2 = 500$$

$$\frac{dR}{dt} = \frac{10}{500} = \frac{1}{50} \checkmark$$

SECTION B

4) b) The constant term of

$$\left(3x^2 - \frac{1}{2x} \right)^9$$

$$U_{r+1} = {}^9C_r (3x^2)^{9-r} \left(-\frac{1}{2x} \right)^r \checkmark$$

$$= A (x^2)^{9-r} (x^{-1})^r$$

$$= A x^{18-2r} \cdot x^{-r}$$

$$= A x^{18-3r} \checkmark$$

where A is the num. coeff of U_{r+1}

$$\text{now } 18-3r=0 \quad \therefore r=6 \checkmark$$

$$\therefore U_7 = {}^9C_6 (3x^2)^{9-6} \left(-\frac{1}{2x} \right)^6$$

$$= \frac{9!}{6!3!} (3x^2)^3 \left(-\frac{1}{2x} \right)^6$$

$$= \frac{567}{16} \quad \text{④}$$

$$\text{② Chord PA } y - ap^2 = \frac{p+q}{2} (x - 2ap)$$

will be satisfied by $(0, a)$

$$a - ap^2 = \frac{p+q}{2} (0 - 2ap)$$

$$a - ap^2 = (p+q)(-ap)$$

$$1 - p^2 = -p^2 - pq$$

$$1 = -pq$$

$$\therefore pq = -1 \quad \text{①}$$

$$\text{ii) Midpt } pq = \left(\frac{2ap+2aq}{2}, \frac{ap^2+aq^2}{2} \right)$$

$$= \left(a(p+q), \frac{a}{2}(p^2+q^2) \right)$$

①

$$4b)(iii) x = a(p+q)$$

$$p+q = \frac{x}{a}$$

$$pq = -1$$

$$y = \frac{a}{2}(p^2+q^2)$$

$$\frac{2y}{a} = (p+q)^2 - 2pq$$

$$\frac{2y}{a} = \left(\frac{x}{a} \right)^2 - 2$$

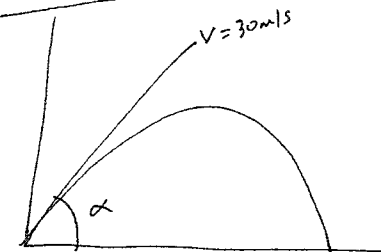
$$2y = \frac{x^2}{a} - 2a$$

$$2ay = x^2 - 2$$

$$x^2 = 2ay + 2$$

$$x^2 = 2(ay+1)$$

Question 5



$$a) \dot{x} = 0$$

$$x = t + c_1 \quad \text{but when } t=0$$

$$\dot{x} = V \cos \alpha$$

$$\therefore c_1 = V \cos \alpha$$

$$\therefore \dot{x} = V \cos \alpha$$

$$= 30 \cos \alpha$$

$$x = 30t \cos \alpha + c_2$$

$$\text{but when } t=0$$

$$\therefore c_2 = 0$$

$$\therefore x = 30t \cos \alpha$$

$$b) \ddot{y} = -g$$

$$= -10$$

$$\dot{y} = -10t + c_3$$

$$\text{now when } t=0 \quad \dot{y} = V \sin \alpha$$

$$= 30 \sin \alpha$$

$$\therefore c_3 = 30 \sin \alpha$$

$$\dot{y} = -10t + 30 \sin \alpha$$

$$y = -5t^2 + 30t \sin \alpha + c_4$$

$$\text{when } t=0 \quad y=0 \therefore$$

$$\therefore y = -5t^2 + 30t \sin \alpha$$

$$\begin{aligned}
 & -5t^2 + 30t \sin \alpha \\
 & = +t(30 \sin \alpha - 5t) \\
 & 0 \Rightarrow 0 = 30 \sin \alpha - 5t \\
 & 5t = 30 \sin \alpha \\
 & t = 6 \sin \alpha
 \end{aligned}$$

$$\begin{aligned}
 & \text{at } t = 6 \sin \alpha \\
 & x = (30 \cos \alpha)(6 \sin \alpha) \\
 & = 2(90 \cos \alpha \cdot \sin \alpha) \\
 & = 90 \sin 2\alpha \quad (2)
 \end{aligned}$$

max dist, x , $\pi \sim$
 up can reach will
 be when $\sin 2\alpha = 1$
 $\therefore$ the max dist is 90m.

(1)

$$\begin{aligned}
 & = 30t \cos \alpha \\
 & = \frac{x}{30 \cos \alpha} \\
 & = -5 \left(\frac{x}{30 \cos \alpha} \right)^2 + 30 \left(\frac{x}{30 \cos \alpha} \right) \sin \alpha \\
 & \text{now find } \alpha \text{ when } y=0, x=60 \\
 & = -5 \left(\frac{60}{30 \cos \alpha} \right)^2 + 30 \left(\frac{60}{30 \cos \alpha} \right) \sin \alpha \\
 & = -5 \left(\frac{4}{\cos^2 \alpha} \right) + 60 \tan \alpha
 \end{aligned}$$

$$\begin{aligned}
 0 &= -20(\tan^2 \alpha + 1) + 60 \tan \alpha \\
 &= \tan^2 \alpha - 3 \tan \alpha + 1
 \end{aligned}$$

$$\begin{aligned}
 \tan \alpha &= \frac{3 \pm \sqrt{9-4(1)(1)}}{2} \\
 &= \frac{3 \pm \sqrt{5}}{2}
 \end{aligned}$$

$$\alpha = 69^\circ 05', 20^\circ 54'$$

(OR)

$$x = 30t \cos \alpha$$

$$60 = 30t \cos \alpha$$

$$2 = t \cos \alpha$$

$$t = \frac{2}{\cos \alpha} \quad \checkmark$$

$$\frac{2}{\cos \alpha} = 6 \sin \alpha$$

$$2 = 6 \sin \alpha \cos \alpha$$

$$2 = 3(2 \sin \alpha \cos \alpha)$$

$$\frac{2}{3} = \sin 2\alpha$$

$$2\alpha = \sin^{-1} \left(\frac{2}{3} \right)$$

$$2\alpha = 41.8, 138.2$$

$$\alpha = \frac{1}{2} \sin^{-1} \left(\frac{2}{3} \right)$$

$$\alpha = 20^\circ 54', 69^\circ 05'$$

$$V = 40 \text{ m/s} \quad \alpha = 0 \quad g = 10$$

$$y = -\frac{1}{2}gt^2 + Vt \sin \alpha$$

$$y = -\frac{1}{2}(10)t^2 + 40t \sin 0$$

$$y = -5t^2$$

$$\text{now when } t=0 \quad y=5$$

$$y = -5t^2 + 5$$

$$x = Vt \cos \alpha$$

$$x = 40t \cos 0$$

$$x = 40t$$

now when $y=0$ the
 pumped water reaches
 the ground

$$0 = -5t^2 + 5$$

$$t = \pm 1 \quad (t \geq 0)$$

$$\therefore t = 1$$

$$\text{at } t=1 \quad x = 40$$

$\therefore$ No water will hit the
 ground 40m away.

It will not reach
 the fire.

i) Show that $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^2 x \, dx = \frac{\pi}{8} - \frac{1}{4}$

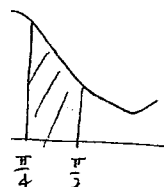
$$\frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\cos 2x + 1) dx = \frac{1}{2} \left[\frac{1}{2} \sin 2x + x \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \frac{1}{2} \left[\left(\frac{1}{2} \sin \pi + \frac{\pi}{2} \right) - \left(\frac{1}{2} \sin \frac{\pi}{2} + \frac{\pi}{4} \right) \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{2} - \frac{1}{2} - \frac{\pi}{4} \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{2} \right]$$

$$= \frac{\pi}{8} - \frac{1}{4} \text{ units}^2 \quad (2)$$



$$V = \pi \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2 + \cos x)^2 dx \quad (1)$$

$$= \pi \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (4 + 4\cos x + \cos^2 x) dx$$

$$= \pi \left[4x + 4\sin x + \int \cos^2 x \, dx \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \pi \left[4x + 4\sin x + \int \cos(2x+1) dx \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \pi \left[4x + 4\sin x \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} + \pi \left[\frac{\pi}{8} - \frac{1}{4} \right] \therefore \text{The amplitude} = 3$$

$$= \pi \left[\left[\frac{4\pi}{2} + 4\sin \frac{\pi}{2} \right] - \left[\frac{4\pi}{4} + 4\sin \frac{\pi}{4} \right] + \left[\frac{\pi}{8} - \frac{1}{4} \right] \right]$$

=

$$= \pi \left[\frac{9\pi}{8} + \frac{15}{4} + 2\sqrt{2} \right] \text{ units}^3$$

6b) i) $v^2 = 16x - 4x^2 + 20$

$$a = \frac{d}{dx} \left[\frac{1}{2} v^2 \right]$$

$$= \frac{d}{dx} [8x - 2x^2 + 10]$$

$$a = 8 - 4x$$

$$a = 4(2-x) = -4(x-2) \quad (2)$$

ii) centre of motion is $x=2$

iii) when $v=0$

$$0 = 16x - 4x^2 + 20$$

$$= -4(x^2 - 4x - 5)$$

$$= -4(x-5)(x+1) \quad (2)$$

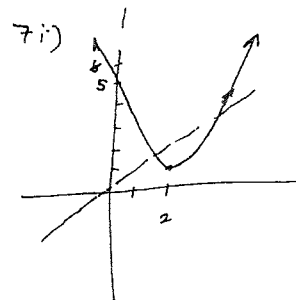
$$x=5 \text{ or } x=-1$$

(iv) $n^2 = 4 \therefore n = 2 \ (n > 0) \quad (1)$

v) Max speed when $x=1$
(centre of motion)

$$v^2 = 4(3+2-1)$$

$$= \pm 4 \quad (2)$$



$$y = (x-2)^2 + 1$$

(1)

v) At $x=a$ $a < 2$

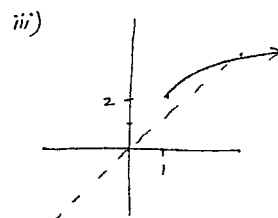
$$f(a) = (a-2)^2 + 1 = A$$

$$f^{-1}(f(a)) = f^{-1}(A)$$

$$= 2 + \sqrt{A-1}$$

$$= 2 + \sqrt{(a-2)^2 + 1 - 1}$$

ii) $x \geq 2$ for $f^{-1}(x) \quad (1)$



(1.5)

1.5

$$= \frac{f(a)}{f(a)}$$

$$= 2 + \sqrt{(a-2)^2} \text{ since } a < 2$$

$$= 2 + 2 - a$$

$$= 4 - a \quad (2)$$

$$\therefore f^{-1}(f(a)) = 4 - a$$

iv) domain of inverse fn

$$\text{is } x \geq 1 \quad (1)$$

vi) Since the inverse fn is the reflection of $f(x)$ about the line $y=x$, pts of int. occur when $y=x$

7a) $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$

from $y = (x-2)^2 + 1$ & $y=x$

$$n(1+x)^{n-1} = {}^nC_1 + 2{}^nC_2 x + \dots + n{}^nC_n x^{n-1} \quad (1)$$

$$x=1$$

$$n \cdot 2^{n-1} = {}^nC_1 + 2{}^nC_2 + \dots + n{}^nC_n \quad (2)$$

$$= \sum_{r=1}^n r \cdot {}^nC_r \quad (1)$$

$$x = (x-2)^2 + 1$$

$$x = (x^2 - 4x + 4) + 1$$

$$x = x^2 - 4x + 5$$

$$0 = x^2 - 5x + 5$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$